

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Fundamentals

Week 1 Day 1 AM



OUTLINE

- ① Sets and Real Numbers
- ② Absolute Value and Distance
- ③ Algebraic Expressions
- ④ Summary
- ⑤ Fractional Expressions [Class Participation Assignment]
- ⑥ Exponentials and Radicals [Class Participation Assignment]

SETS AND ELEMENTS

DEFINITION:

A **set** is a collection of distinct objects. These objects are called the **elements** of the set.

E.g.:

- The set of all students in SUTD.
- $\{1, 3, 4, 13\}$.
- $\{a, s, 7, \pi, \text{SUTD}\}$.
- $\{\text{the Sun, the Moon, Earth}\}$.
- $\{0, 2, 4, 6, 8, \dots\}$.

The curly brackets $\{ \}$ are the standard notations used to represent a set. The things inside the curly brackets are the elements of the set.

We use three dots " $\dots$ " to abbreviate some elements when the pattern is clear.

The order of elements listed in a set is not important. For example, the two sets $\{1, 2, 3\}$ and $\{2, 1, 3\}$ are the same.

EXAMPLES OF SETS

Some common sets/notations:

- $\mathbb{N} := \{0, 1, 2, \dots\}$, the set of **natural numbers** including 0.
- $\mathbb{N}^* := \{1, 2, 3, \dots\}$, the set of natural numbers excluding 0.
- $\mathbb{Z} := \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, the set of all **integers**.
- $\mathbb{Q} :=$ the set of all **rational numbers**, i.e. numbers that can be expressed as $\frac{m}{n}$ where m, n are integers and $n \neq 0$.
- $\mathbb{R} :=$ the set of all **real numbers**.
- $\mathbb{R}^2 := \{(x, y) : x, y \text{ are real numbers}\}$, the set of all **ordered pairs** of real numbers.
- $\mathbb{C} :=$ the set of all **complex numbers**.
- $\emptyset$ or $\{\}$:= the **empty set** or **null set**, i.e. the set with no elements.

⚠ Do not confuse 0 with $\emptyset$, they are different. The former is a number, the latter is a set containing nothing.

REAL NUMBERS

DEFINITION:

Real numbers that cannot be expressed as a ratio of integers are called ***irrational numbers***.

Every real number has a decimal representation. If the number is rational, then its corresponding decimal is repeating eventually. (The bar on top of the digits indicates that the sequence of digits repeats forever.)

E.g.:

- $\frac{1}{4} = 0.25 = 0.25000 \dots = 0.25\overline{0}$.
- $\frac{2642}{825} = 3.20242424 \dots = 3.20\overline{24}$.

Qn: Given a decimal representation that is repeating eventually, do you know how to rewrite it as a ratio of two integers? Hint: check out the e.g. given on page 3 of the textbook.

REAL NUMBERS

If a number is irrational, then its decimal representation is nonrepeating.

E.g.:

- $\pi = 3.141592653589793 \dots$
- $\sqrt{2} = 1.41421356 \dots$
- $e = 2.71828182845904523 \dots$
- $\ln 3 = 1.0986122 \dots$

We can write $\pi \approx 3.14159$, where the symbol $\approx$ means "is approximately equal to".

⚠ The notation $\sqrt{x}$ means $+\sqrt{x}$ and not $\pm\sqrt{x}$, which is the same convention as numbers: 7 means $+7$ and not ± 7 .

Fun facts:

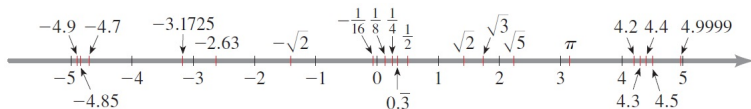
- There are as many natural numbers as integers.
- There are as many rational numbers as integers.
- There are way more irrational numbers than the rational numbers. If we represent the real numbers as a line and we randomly cut it somewhere, most likely we'll hit an irrational number.

THE REAL LINE

The real numbers can be represented by points on a line, called the *real line*.

Each positive number x is represented by the point on the line a distance of x units to the right of the origin $= 0$.

Each negative number $-x$ is represented by the point x units to the left of the origin.



Fun facts:

- Rational numbers are dense in the real line, i.e. there's always a rational number between any two distinct real numbers. The same is true for irrational numbers.
- There are no gaps/holes in the real line.
- There's no such thing as the real number next to a given real number. For instance, what is the real number next to 3?

ELEMENTS, SUBSETS

NOTATION:

The symbol $\in$ means 'is an element of', whereas $\notin$ means 'is not an element of'.

In general, writing a slash / over a symbol means 'not'.

E.g.:

- $1 \in \{1, 3, 4, 13\}.$
- $0.1 \notin \mathbb{Z}.$
- $-7 \in \mathbb{Z}.$
- $\pi \notin \mathbb{Q}.$

DEFINITION:

A set X is a **subset** of a set Y if every element of X is also an element of Y . We write this as $X \subseteq Y$.

E.g.:

- $\{1, 3, 4, 13\} \subseteq \mathbb{Z}.$
- $\mathbb{R} \not\subseteq \mathbb{Z}.$
- $\mathbb{Q} \subseteq \mathbb{R}.$
- $\mathbb{Z} \notin \mathbb{R}$, but $\mathbb{Z} \subseteq \mathbb{R}.$

SETS EQUALITY

DEFINITION:

Two sets, X and Y , are **equal** precisely when $X \subseteq Y$ and $Y \subseteq X$. We write this as $X = Y$.

This definition also implies that $X = Y$ means X and Y have exactly the same elements.

⚠ This definition is handy, especially when two sets are very complicated and you can't eyeball that they have the same elements. To prove that two sets A, B are equal: first show that $A \subseteq B$, then show that $B \subseteq A$, and so $A = B$.

⚠ The subset inclusion $A \subseteq B$ is not strict, i.e. it's possible that $A = B$. For strict subset inclusions, use $A \subsetneq B$.

ACTIVITY 1

Given $A = \{-\sqrt{2}, \frac{1}{3}, 2, \pi\}$, $B = \{\}$ and $C = \{2, 3, 5, 7, 11\}$, fill in the blanks with $\in$, $\notin$, $\subseteq$, $\not\subseteq$, or $=$.

(a) $2\pi \square A$

(b) $0 \square B$

(c) $7 \square C$

(d) $\{\pi\} \square A$

(e) $B \square A$

(f) $\{3, 5, 7, 9, 11\} \square C$

(g) $\{11, 5, 3, 7, 2\} \square C$

ACTIVITY 1: SOLUTION

$$A = \left\{-\sqrt{2}, \frac{1}{3}, 2, \pi\right\}, B = \{\} \text{ and } C = \{2, 3, 5, 7, 11\}.$$

- (a) $2\pi \notin A$, though $2 \in A$ and $\pi \in A$.
- (b) $0 \notin B$, because $B = \{\}$ is the empty set that doesn't contain any elements.
- (c) $7 \in C$.
- (d) $\{\pi\} \subseteq A$ or $\{\pi\} \notin A$, but $\pi \in A$.
- (e) $B \subseteq A$, because B doesn't have any element that doesn't belong to A . In general, $\{\}$ is a subset of any set.
- (f) $\{3, 5, 7, 9, 11\} \not\subseteq C$ or $\{3, 5, 7, 9, 11\} \notin C$.
- (g) $\{11, 5, 3, 7, 2\} = C$, because the order of listing the elements in a set doesn't matter.

SET-BUILDER NOTATION

NOTATION:

The notations

$$S := \{x \in X : \text{some properties involving } x\}$$

and

$$S := \{x \in X \mid \text{some properties involving } x\}$$

means S is the set containing exactly those $x \in X$ that satisfy all the properties stated after the colon $:$ or vertical line $|$.

E.g.:

$$\mathbb{Q} = \left\{ x \in \mathbb{R} \mid x = \frac{m}{n}, \text{ where } m, n \in \mathbb{Z} \text{ and } n \neq 0 \right\},$$

i.e. $\mathbb{Q}$ is the set of **real numbers** x that can be written as $x = \frac{m}{n}$, for some integers $m, n \neq 0$.

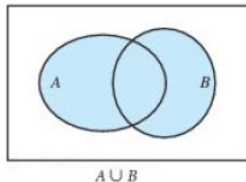
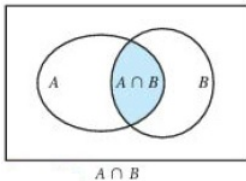
⚠ Make sure you understand this notation, you will see it a lot in the **future** as it's a concise way of defining a set.

UNIONS AND INTERSECTIONS

DEFINITION:

The **union** of sets A and B , denoted by $A \cup B$, is the set whose elements are **either** in A **or** B (or both). The **intersection** of sets A and B , denoted by $A \cap B$, is the set whose elements are in **both** A **and** B .

These notions can be represented on a **Venn diagram**.



E.g.: Let $A := \{1, 2, 3\}$ and $B := \{2, 4, 6\}$. Then $A \cup B = \{1, 2, 3, 4, 6\}$ and $A \cap B = \{2\}$.

COMPLEMENTS

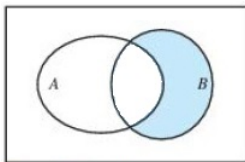
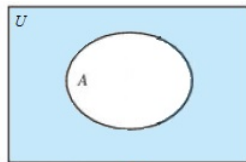
DEFINITION:

Let A, B be two sets. The **complement** of A **relative to** B , denoted by $B \setminus A$, is the set of elements of B that are not in A , i.e.

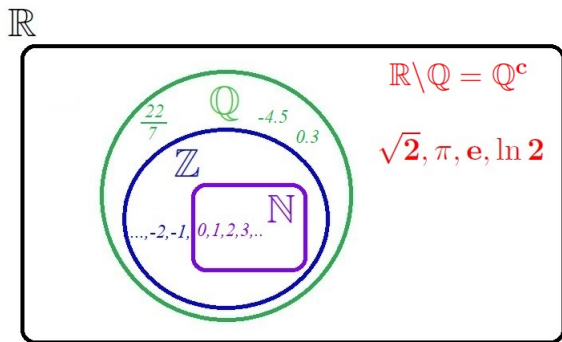
$$B \setminus A := \{x \in B \mid x \notin A\}.$$

DEFINITION:

Let $A \subseteq U$ (U is called the universal set, it should be clear in the given context), the **complement** of A , denoted by A^c or A' , is the set of elements of U that are not in A , i.e. $A^c = U \setminus A$.

 $B \setminus A$  A^c

VENN DIAGRAM FOR REAL NUMBERS



Note that the set of **irrational numbers** is the complement of the set of **rational numbers** $\mathbb{Q}$.

INTERVALS

Notation	Set description	Graph
(a, b)	$\{x \mid a < x < b\}$	
$[a, b]$	$\{x \mid a \leq x \leq b\}$	
$[a, b)$	$\{x \mid a \leq x < b\}$	
$(a, b]$	$\{x \mid a < x \leq b\}$	
(a, ∞)	$\{x \mid a < x\}$	
$[a, \infty)$	$\{x \mid a \leq x\}$	
$(-\infty, b)$	$\{x \mid x < b\}$	
$(-\infty, b]$	$\{x \mid x \leq b\}$	
$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	

The symbol ∞ (“infinity”) does not stand for a real number, that’s why it’s always excluded (in the context of real numbers).

ACTIVITY 2

Given $A = \{1, 2, 3, 4, 5, 6, 7\}$, $B = \{2, 4, 6, 8\}$ and $C = \{7, 8, 9, 10\}$, find

- (a) $A \cup B$
- (b) $A \cap B$
- (c) $A \setminus B$
- (d) $B \setminus A$
- (e) $(A \cup B) \cup C$
- (f) $(A \cap B) \cap C$
- (g) $(A \cup B) \cap C$
- (h) $(A \cap B) \cup C$
- (i) $(A \cup B) \setminus C$

- (j) Rewrite $A = \{x \in \square \mid \square \leq x \leq \square\} = \square \cap [\square, \square]$.

ACTIVITY 2: SOLUTION

$$A = \{1, 2, 3, 4, 5, 6, 7\}, B = \{2, 4, 6, 8\} \text{ and } C = \{7, 8, 9, 10\}.$$

$$\begin{aligned} \text{(a)} \quad A \cup B \\ = \{1, 2, 3, 4, 5, 6, 7, 8\}. \end{aligned}$$

$$\text{(b)} \quad A \cap B = \{2, 4, 6\}.$$

$$\text{(c)} \quad A \setminus B = \{1, 3, 5, 7\}.$$

$$\text{(d)} \quad B \setminus A = \{8\}.$$

$$\begin{aligned} \text{(e)} \quad (A \cup B) \cup C \\ = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}. \end{aligned}$$

$$\text{(f)} \quad (A \cap B) \cap C = \emptyset.$$

$$\text{(g)} \quad (A \cup B) \cap C = \{7, 8\}.$$

$$\begin{aligned} \text{(h)} \quad (A \cap B) \cup C \\ = \{2, 4, 6, 7, 8, 9, 10\}. \end{aligned}$$

$$\begin{aligned} \text{(i)} \quad (A \cup B) \setminus C \\ = \{1, 2, 3, 4, 5, 6\}. \end{aligned}$$

$$\text{(j)} \quad A = \{x \in \mathbb{N} \mid 1 \leq x \leq 7\} = \mathbb{N} \cap [1, 7].$$

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ABSOLUTE VALUE

Intuitively, the absolute value of a real number x , denoted by $|x|$, is the **distance** from x to 0 on the real number line. Since distance is always greater than or equal to 0, we have $|x| \geq 0$. More formally,

DEFINITION:

For any real number x , the **absolute value** of x is

$$|x| := \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

E.g.:

- $|2| = 2$.
- $|-3| = -(-3) = 3$.
- $|0| = 0$.
- $|e - \pi| = -(e - \pi) = \pi - e$, because $e = 2.718 \dots < 3.141 \dots = \pi$, so $e - \pi < 0$.

PROPERTIES OF ABSOLUTE VALUE

PROPERTIES OF ABSOLUTE VALUE

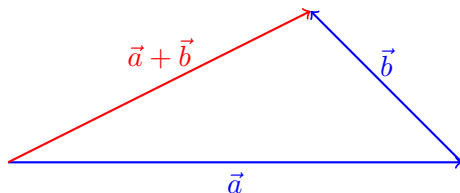
Property	Example	Description
1. $ a \geq 0$	$ -3 = 3 \geq 0$	The absolute value of a number is always positive or zero.
2. $ a = -a $	$ 5 = -5 $	A number and its negative have the same absolute value.
3. $ ab = a b $	$ -2 \cdot 5 = -2 5 $	The absolute value of a product is the product of the absolute values.
4. $\left \frac{a}{b}\right = \frac{ a }{ b }$	$\left \frac{12}{-3}\right = \frac{ 12 }{ -3 }$	The absolute value of a quotient is the quotient of the absolute values.
5. $ a + b \leq a + b $	$ -3 + 5 \leq -3 + 5 $	Triangle Inequality

6. $|a| \geq a$ the absolute value of a number is at least itself

7. $||a| - |b|| \leq |a - b| \leq |a| + |b|$ extended triangle inequality

⚠ Note that for multiplications and divisions, you can split the absolute value to each term (we say that the absolute value commutes with multiplications and divisions). However, you can't do this for additions and subtractions, the best we can do is the triangle inequality.

TRIANGLE INEQUALITY



Any **side** of a triangle is shorter than or equal to the sum of the **two other sides**.

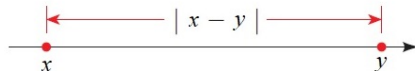
$$|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$$

DISTANCE

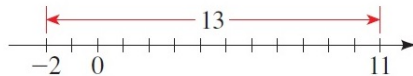
DEFINITION:

For any two real numbers x, y , the **distance** between them on the real line is

$$d(x, y) := |x - y|.$$



E.g.: $d(-2, 11) = |-2 - 11| = |-13| = 13.$



ACTIVITY 3

(a) Simplify the following.

i) $|\sqrt{5} - 5|$

ii) $||-5| - |-12||$

iii) $a + b + |a - b|$, where $a < b$.

(b) By using the definition of distance, solve for all x 's that satisfy

$$|x - 3| \leq 1.$$

Graph your solution.

(c) Let $\max\{a, b\}$ denote the maximum and $\min\{a, b\}$ denote the minimum of the real numbers a and b . For example, $\max\{2, 5\} = 5$ and $\min\{1, -2\} = -2$.

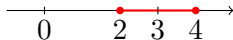
i) Prove that $\max\{a, b\} = \frac{a+b+|a-b|}{2}$.

ii) Prove that $\min\{a, b\} = \frac{a+b-|a-b|}{2}$.

Hint: Use part (a)iii).

ACTIVITY 3: SOLUTION

- (a) i) $|\sqrt{5} - 5| = 5 - \sqrt{5}$, because $25 > 5 \Leftrightarrow \sqrt{25} = 5 > \sqrt{5}$.
- ii) $||-5| - |-12|| = |5 - 12| = |-7| = 7$.
- iii) Since $a < b$, so $|a - b| = b - a$ and thus
 $a + b + |a - b| = a + b + b - a = 2b$.
- (b) The x 's are real numbers that are at a distance ≤ 1 from 3,
i.e. $-1 \leq x - 3 \leq 1 \iff 2 \leq x \leq 4$.



ACTIVITY 3: SOLUTION

(c)

i) If $a < b$, then $\max\{a, b\} = b$. From (a)iii), we have $\frac{a+b+|a-b|}{2} = \frac{2b}{2} = b = \max\{a, b\}$. The case of $a > b$ is similar by symmetry. The case $a = b$ is trivial.

ii) If $a < b$, then $\min\{a, b\} = a$ and $|a - b| = b - a$. So $\frac{a+b-|a-b|}{2} = \frac{a+b-(b-a)}{2} = \frac{2a}{2} = a = \min\{a, b\}$. The case of $a > b$ is similar by symmetry. The case $a = b$ is trivial.

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ALGEBRAIC EXPRESSIONS

DEFINITION:

A **variable** is a letter that represents any number from a given set. Combining some variables using $+$, $-$, $\times$, $\div$, powers, and roots produces an **algebraic expression**.

E.g.:

- $2x^2 - 3x + 4$ (x is a variable, 2 and -3 are **coefficients** of x^2 and x respectively, 4 is a constant.)
- $\sqrt{x} + 10$
- $\frac{y - 2z}{y^2 + 4}$

POLYNOMIAL EXPRESSIONS

DEFINITION:

A **polynomial expression** is an algebraic expression of the form

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0,$$

where $a_n, a_{n-1}, \dots, a_1, a_0$ are real numbers, and n is a non-negative integer. If $a_n \neq 0$, then the polynomial has **degree n** . The expressions $a_k x^k$ that make up the polynomial are called the **terms** of the polynomial.

Remarks:

1. Since addition is commutative, a polynomial remains the same if we reorder the terms.
2. The degree of a polynomial is the highest power of the variable that appears in the polynomial.

EXAMPLES OF POLYNOMIAL EXPRESSIONS

Polynomial	Type	Terms	Degree
$2x^2 - 3x + 4$	trinomial	$2x^2, -3x, 4$	2
$x^8 + 5x$	binomial	$x^8, 5x$	8
$8 - x + x^2 - \frac{1}{2}x^3$	four terms	$-\frac{1}{2}x^3, x^2, -x, 3$	3
$5x + 1$	binomial	$5x, 1$	1
$9x^5$	monomial	$9x^5$	5
6	monomial	6	0

The type monomial, binomial and trinomial are special terminologies given to polynomials with one, two and three terms respectively.

ADDING AND SUBTRACTING ALGEBRAIC EXPRESSIONS

We add and subtract algebraic expressions using the properties of real numbers discussed in the appendix. The idea is to combine **like terms** (i.e. terms with the same variable raised to the same powers) using the distributive property.

E.g.:

- $\sqrt{x} + 2x^2 - 3\sqrt{x} + 5x^2 = (\sqrt{x} - 3\sqrt{x}) + (2x^2 + 5x^2) = (1 - 3)\sqrt{x} + (2 + 5)x^2 = -2\sqrt{x} + 7x^2.$
- $x - y - (x + y) = x - y - x - y = x - x - y - y = -2y.$

MULTIPLYING ALGEBRAIC EXPRESSIONS

Similarly, to multiply algebraic expressions, we use the properties of real numbers and the distributive property repeatedly.

E.g.:

- $\left(2x + \frac{1}{x}\right) \left(3x - \frac{5}{x}\right) = 2x \left(3x - \frac{5}{x}\right) + \frac{1}{x} \left(3x - \frac{5}{x}\right) =$
 $6x^2 - 10 + 3 - \frac{5}{x^2} = 6x^2 - 7 - \frac{5}{x^2}.$
- $(2x + 3)(x^2 - 5x + 4) = 2x(x^2 - 5x + 4) + 3(x^2 - 5x + 4) =$
 $2x^3 - 10x^2 + 8x + 3x^2 - 15x + 12 = 2x^3 - 7x^2 - 7x + 12.$

SPECIAL PRODUCT/FACTORING FORMULAS

SPECIAL PRODUCT FORMULAS

If A and B are any real numbers or algebraic expressions, then

- | | |
|--|----------------------------------|
| 1. $(A + B)(A - B) = A^2 - B^2$ | Sum and difference of same terms |
| 2. $(A + B)^2 = A^2 + 2AB + B^2$ | Square of a sum |
| 3. $(A - B)^2 = A^2 - 2AB + B^2$ | Square of a difference |
| 4. $(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$ | Cube of a sum |
| 5. $(A - B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$ | Cube of a difference |

You don't need to memorize them, just expand them using the distributive property, for instance

$$\begin{aligned}(A + B)^2 &= (A + B)(A + B) = A(A + B) + B(A + B) \\ &= A^2 + AB + BA + B^2 = A^2 + 2AB + B^2.\end{aligned}$$

But it's important to be able to recognize and factorize (going from the RHS to the LHS) expressions 1, 2, and 3.

SPECIAL PRODUCT FORMULAS

You need to be flexible in applying the formulas, for instance

$$(x^2 + y^3)^2 = (x^2)^2 + 2(x^2)(y^3) + (y^3)^2$$


$$(A + B)^2 = A^2 + 2AB + B^2$$

Also, the identity $(A + B)(A - B) = A^2 - B^2$ is useful to "remove" radicals:

$$(\sqrt{x} + \sqrt{y})(\sqrt{x} - \sqrt{y}) = (\sqrt{x})^2 - (\sqrt{y})^2 = x - y.$$

FACTORING COMMON FACTORS

If there's a common factor among the terms, we can factor it out (basically it's the reverse process of the distributive property).

E.g.:



$$8x^3y^2 - 8x^2y^3 + 2xy^4 = 2xy^2(4x^2 - 4xy + y^2) = 2xy^2(2x - y)^2.$$



$$\begin{aligned} & (2x + 4)(x - 3) - (5x^2 - 1)(x - 3) \\ &= (2x + 4 - (5x^2 - 1))(x - 3) \\ &= (2x + 4 - 5x^2 + 1)(x - 3) = (-5x^2 + 2x + 5)(x - 3). \end{aligned}$$

FACTORING $ax^2 + bx + c$

Suppose we want to factorize a quadratic $ax^2 + bx + c$ as $(px + r)(qx + s)$, then we must have

$$ax^2 + bx + c = (px + r)(qx + s) = pqx^2 + (ps + qr)x + rs.$$

If a, b, c are **integers** and we suspect that the roots of the quadratic are **rational** numbers (not always true), then

- p, q, r, s are integers as well,
- $pq = a$,
- $rs = c$,
- $ps + qr = b$.

Thus, this gives us a strategy to factorize quadratics (with integer coefficients) by narrowing it down to trying all possible divisors (including the negative ones) of a and c .

FACTORING $ax^2 + bx + c$

E.g.: Suppose we suspect that $6x^2 + 7x - 5 = (px + r)(qx + s)$ has rational roots, then we'll try $pq = 6$ and $rs = -5$ such that $ps + qr = 7$.

Case 1: $p = 1, q = 6, r = -1, s = 5$, then we have
 $(x - 1)(6x + 5) = 6x^2 - x - 5 \neq 6x^2 + 7x - 5$.

Case 2: $p = 1, q = 6, r = 1, s = -5$, then we have
 $(x + 1)(6x - 5) = 6x^2 + x - 5 \neq 6x^2 + 7x - 5$.

Case 3: $p = 2, q = 3, r = -1, s = 5$, then we have
 $(2x - 1)(3x + 5) = 6x^2 + 7x - 5 = 6x^2 + 7x - 5$, bingo!

⚠ If the quadratic has irrational roots, you'll not get the factorization using this approach. A bulletproof method is to use the quadratic roots formula (see the appendix of today's PM slides):

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

FACTORING $x^2 + bx + c$

Note that in the special case $a = 1$, the previous method simplifies to

$$x^2 + bx + c = (x + r)(x + s) = x + (r + s)x + rs,$$

with

- $rs = c$,
- $r + s = b$.

E.g.: Suppose we suspect that $x^2 - 6x + 5 = (x + r)(x + s)$ has rational roots, then we'll try $rs = 5$ and $r + s = -6$. $r = -5$, $s = -1$ satisfy the conditions and thus

$$x^2 - 6x + 5 = (x - 5)(x - 1).$$

E.g.: Suppose we suspect that $x^2 + 7x - 5 = (x + r)(x + s)$ has rational roots, then we'll try $rs = -5$ and $r + s = 7$. But $r = 5$, $s = -1$ gives $r + s = 4 \neq 7$, whereas $r = -5$, $s = 1$ gives $r + s = -4 \neq 7$. Since $-5 = (-5) \times 1 = 5 \times (-1)$ are all the possible integer factorizations, we conclude that the roots are irrational.

ACTIVITY 4

Expand the following algebraic expressions:

(a) $(2x + 3y)^2$

(b) $(2y + 5)(2y - 5)$

(c) $(\sqrt{y} + \sqrt{2})(\sqrt{y} - \sqrt{2})$

(d) $(1 - 2r)^3$

Factorize the following algebraic expressions:

(e) $-2x^3 + x$

(f) $8x^2 - 14x - 15$

(g) $2x^2y - 6xy^2 + 3xy$

(h) $x^4 + x^2 - 2$

ACTIVITY 4: SOLUTION

$$(a) \quad (2x + 3y)^2 = (2x + 3y)(2x + 3y) = 4x^2 + 6xy + 6xy + 9y^2 = 4x^2 + 12xy + 9y^2.$$

$$(b) \quad (2y + 5)(2y - 5) = (2y)^2 - 5^2 = 4y^2 - 25.$$

$$(c) \quad (\sqrt{y} + \sqrt{2}) (\sqrt{y} - \sqrt{2}) = (\sqrt{y})^2 - (\sqrt{2})^2 = y - 2.$$

$$(d) \quad (1 - 2r)^3 = (1 - 2r)(1 - 2r)(1 - 2r) = (1 - 2r - 2r + 4r^2)(1 - 2r) = (1 - 4r + 4r^2)(1 - 2r) = 1 - 4r + 4r^2 - 2r + 8r^2 - 8r^3 = 1 - 6r + 12r^2 - 8r^3.$$

$$(e) \quad -2x^3 + x = x(-2x^2 + 1) = x(1 - 2x^2) = x \left(1^2 - (\sqrt{2}x)^2 \right) = x(1 - \sqrt{2}x)(1 + \sqrt{2}x).$$

$$(f) \quad 8x^2 - 14x - 15 = (px + r)(qx + s), \text{ where } pq = 8, rs = -15 \text{ and } ps + qr = -14. \text{ Try } p = 4, q = 2, r = 3, s = -5, \text{ we do get } 8x^2 - 14x - 15 = (4x + 3)(2x - 5).$$

ACTIVITY 4: SOLUTION

(g) $2x^2y - 6xy^2 + 3xy = xy(2x - 6y + 3).$

(h) $x^4 + x^2 - 2 = y^2 + y - 2$ if we let $y = x^2$. Then
 $y^2 + y - 2 = (y + r)(y + s)$ with $r + s = 1$, $rs = -2$. Try
 $r = -1$, $s = 2$, we do get $y^2 + y - 2 = (y - 1)(y + 2)$. So
 $x^4 + x^2 - 2 = y^2 + y - 2 = (y - 1)(y + 2) =$
 $(x^2 - 1)(x^2 + 2) = (x + 1)(x - 1)(x^2 + 2).$

OUTLINE

- ① Sets and Real Numbers
- ② Absolute Value and Distance
- ③ Algebraic Expressions
- ④ Summary
- ⑤ Fractional Expressions [Class Participation Assignment]
- ⑥ Exponentials and Radicals [Class Participation Assignment]

SUMMARY

- Sets and Real Numbers
 - sets, elements, subsets.
 - real numbers, irrational numbers, the real line.
 - set-builder notation, unions, intersections, complements.
 - intervals.
- Absolute Value and Distance
- Algebraic Expressions
 - algebraic and polynomial expressions.
 - adding, subtracting and multiplying algebraic expressions.
 - special product formulas.
 - factoring common factors, factoring $ax^2 + bx + c$.
- Fractional Expressions [Class Participation Assignment]
 - simplifying, multiplying, dividing, adding and subtracting.
 - compound fractions.
 - rationalizing.
- Exponentials and Radicals [Class Participation Assignment]
 - properties of real numbers.
 - integer exponents.
 - order of operations.
 - law of exponents.
 - radicals, properties of n th roots.
 - rational exponents.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 1.1, 1.2, 1.3 1.4.

Additional resources:

- A [video](#) on irrationality of e .
- A [video](#) on irrationality of π .
- A [video](#) on irrationality of $\sqrt{2}$.

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FRACTIONAL AND RATIONAL EXPRESSIONS

DEFINITION:

A quotient of two algebraic expressions is called a ***fractional expression***.

E.g.:

$$\frac{1}{2x+1}, \quad \frac{x^3+1}{x-1}, \quad \frac{x}{\sqrt{x^2+1}}, \quad \frac{x+y}{\sqrt{x^2+y^2}}, \quad \frac{x^{\frac{3}{5}}+x+7}{\sqrt{x}-1}.$$

DEFINITION:

A quotient of two **polynomial** expressions is called a ***rational expression***.

E.g.:

$$\frac{1}{2x+1}, \quad \frac{x^3}{x-1}, \quad \frac{x^2+2x+2}{x^2+1}, \quad \frac{x^3-x^2+x-1}{(x^4+2)(x-1)}.$$

SIMPLIFYING, MULTIPLYING AND DIVIDING FRACTIONAL EXPRESSIONS

To simplify fractional expressions, we factor both numerator and denominator and use the following property of fractions:

$$\frac{A\cancel{C}}{B\cancel{C}} = \frac{A}{B}.$$

To multiply fractional expressions, we use the following property of fractions:

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}.$$

To divide fractional expressions, we use the following property of fractions:

$$\frac{\frac{A}{B}}{\frac{C}{D}} = \frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \times \frac{D}{C} = \frac{AD}{BC}.$$

ADDING AND SUBTRACTING FRACTIONAL EXPRESSIONS

To add or subtract fractional expressions, we first find a common denominator and then use the following property of fractions:

$$\frac{A}{C} + \frac{B}{C} = \frac{A+B}{C}, \quad \frac{A}{C} - \frac{B}{C} = \frac{A-B}{C}.$$

The simplest (but not necessarily the most efficient) way to find a common denominator is to just multiply the denominators:

$$\frac{A}{B} + \frac{C}{D} = \frac{AD}{BD} + \frac{BC}{BD} = \frac{AD+BC}{BD},$$

$$\frac{A}{B} - \frac{C}{D} = \frac{AD}{BD} - \frac{BC}{BD} = \frac{AD-BC}{BD}.$$

COMPOUND FRACTIONS

A compound fraction is a fraction in which the numerator, the denominator, or both, are themselves fractional expressions. You can simplify them using the properties of fractions on slide 59.

E.g.:

$$\frac{\frac{x}{y} + 1}{1 - \frac{y}{x}}, \quad \frac{\frac{1}{a+h} - \frac{1}{a}}{h}, \quad \frac{\sqrt{1+x^2} - \frac{x^2}{\sqrt{1+x^2}}}{1+x^2}.$$

Fun fact: A continued fraction is a repeatedly compound fraction:

$$b_0 + \frac{a_1}{b_1 + \frac{a_2}{b_2 + \frac{a_3}{b_3 + \ddots}}}.$$

Mathematicians are able to prove **amazing things** using continued fractions, for instance:

$$\sqrt{2} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \ddots}}}.$$

Find out more [here](#).

RATIONALIZING THE DENOMINATOR OR THE NUMERATOR

If a fraction has a denominator or numerator of the form $A\sqrt{B} + C\sqrt{D}$, we can rationalize it by multiplying both the numerator and denominator by the **conjugate radical** $A\sqrt{B} - C\sqrt{D}$ (flip the sign of the second term only), i.e. multiplying $\frac{A\sqrt{B}-C\sqrt{D}}{A\sqrt{B}-C\sqrt{D}} = 1$ and thus not changing anything. This is mainly due to the identity on slide 34:

$$(A\sqrt{B} + C\sqrt{D})(A\sqrt{B} - C\sqrt{D}) = (A\sqrt{B})^2 - (C\sqrt{D})^2 = A^2B - C^2D.$$

E.g.:

$$\bullet \frac{1}{1+\sqrt{2}} = \frac{1}{1+\sqrt{2}} \times \frac{1-\sqrt{2}}{1-\sqrt{2}} = \frac{1-\sqrt{2}}{1^2-(\sqrt{2})^2} = \frac{1-\sqrt{2}}{-1} = \sqrt{2} - 1.$$

$$\bullet \frac{\sqrt{4+h}-2}{h} = \frac{\sqrt{4+h}-2}{h} \times \frac{\sqrt{4+h}+2}{\sqrt{4+h}+2} = \frac{(\sqrt{4+h})^2-2^2}{h(\sqrt{4+h}+2)} = \frac{4+h-4}{h(\sqrt{4+h}+2)} = \frac{h}{h(\sqrt{4+h}+2)} = \frac{1}{\sqrt{4+h}+2}.$$

COMMON MISTAKES

Don't make the mistake of applying properties of multiplication to the operation of addition:

Correct multiplication property	Common error with addition
$(a \cdot b)^2 = a^2 \cdot b^2$	$(a + b)^2 \neq a^2 + b^2$
$\sqrt{a \cdot b} = \sqrt{a} \sqrt{b} \quad (a, b \geq 0)$	$\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$
$\sqrt{a^2 \cdot b^2} = a \cdot b \quad (a, b \geq 0)$	$\sqrt{a^2 + b^2} \neq a + b$
$\frac{1}{a} \cdot \frac{1}{b} = \frac{1}{a \cdot b}$	$\frac{1}{a} + \frac{1}{b} \neq \frac{1}{a + b}$
$\frac{ab}{a} = b$	$\frac{a + b}{a} \neq b$
$(a \cdot b)^{-1} = a^{-1} \cdot b^{-1}$	$(a + b)^{-1} \neq a^{-1} + b^{-1}$

ACTIVITY 5

(a) Simplify the following.

i) $\frac{x^2-x-2}{x^2-1}$

ii) $\frac{x^2+2x-15}{x^2-25} \cdot \frac{x-5}{x+2}$

iii) $\frac{x+3}{4x^2-9} \div \frac{x^2+7x+12}{2x^2+7x-15}$

iv) $\frac{\frac{x^3}{x+1}}{\frac{x}{x^2+2x+1}}$

v) $\frac{x}{x-4} - \frac{3}{x+6}$

vi) $\frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h}$

(b) Rationalize the numerator of $\frac{\sqrt{x}-\sqrt{y}}{2(x-y)}$.

(c) Rationalize the denominator of $\frac{2}{\sqrt{3}+\sqrt{5}}$

ACTIVITY 5: SOLUTION

$$(a) \quad i) \quad \frac{x^2 - x - 2}{x^2 - 1} = \frac{(x+1)(x-2)}{(x-1)(x+1)} = \frac{x-2}{x-1}.$$

$$ii) \quad \frac{x^2 + 2x - 15}{x^2 - 25} \cdot \frac{x-5}{x+2} = \frac{(x+5)(x-3)}{(x-5)(x+5)} \cdot \frac{x-5}{x+2} = \frac{x-3}{x+2}.$$

$$\begin{aligned} iii) \quad \frac{x+3}{4x^2-9} \div \frac{x^2+7x+12}{2x^2+7x-15} &= \\ \frac{x+3}{(2x-3)(2x+3)} \cdot \frac{2x^2+7x-15}{x^2+7x+12} &= \\ \frac{x+3}{(2x-3)(2x+3)} \cdot \frac{(2x-3)(x+5)}{(x+3)(x+4)} &= \frac{x+5}{(2x+3)(x+4)}. \end{aligned}$$

$$iv) \quad \frac{\frac{x^3}{x+1}}{\frac{x}{x^2+2x+1}} = \frac{x^3(x^2+2x+1)}{x(x+1)} = \frac{x^3(x+1)^2}{x(x+1)} = x^2(x+1).$$

ACTIVITY 5: SOLUTION

$$\begin{aligned} \text{(a) v)} \quad \frac{x}{x-4} - \frac{3}{x+6} &= \frac{x(x+6) - 3(x-4)}{(x-4)(x+6)} = \\ \frac{x^2 + 6x - 3x + 12}{(x-4)(x+6)} &= \frac{x^2 + 3x + 12}{(x-4)(x+6)}. \end{aligned}$$

$$\begin{aligned} \text{vi)} \quad \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h} &= \frac{\frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x}\sqrt{x+h}}}{h} = \frac{\sqrt{x} - \sqrt{x+h}}{h\sqrt{x}\sqrt{x+h}} = \\ \frac{\sqrt{x} - \sqrt{x+h}}{h\sqrt{x}\sqrt{x+h}} \cdot \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} &= \\ \frac{x - (x+h)}{h\sqrt{x}\sqrt{x+h}(\sqrt{x} + \sqrt{x+h})} &= \\ \frac{-1}{\sqrt{x}\sqrt{x+h}(\sqrt{x} + \sqrt{x+h})}. \end{aligned}$$

ACTIVITY 5: SOLUTION

$$(b) \frac{\sqrt{x} - \sqrt{y}}{2(x - y)} = \frac{\sqrt{x} - \sqrt{y}}{2(x - y)} \cdot \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}} = \frac{x - y}{2(x - y)(\sqrt{x} + \sqrt{y})} = \frac{1}{2(\sqrt{x} + \sqrt{y})}.$$

$$(c) \frac{2}{\sqrt{3} + \sqrt{5}} = \frac{2}{\sqrt{3} + \sqrt{5}} \cdot \frac{\sqrt{3} - \sqrt{5}}{\sqrt{3} - \sqrt{5}} = \frac{2(\sqrt{3} - \sqrt{5})}{3 - 5} = \sqrt{5} - \sqrt{3}.$$

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PROPERTIES OF REAL NUMBERS

PROPERTIES OF REAL NUMBERS

Property	Example	Description
Commutative Properties		
$a + b = b + a$	$7 + 3 = 3 + 7$	When we add two numbers, order doesn't matter.
$ab = ba$	$3 \cdot 5 = 5 \cdot 3$	When we multiply two numbers, order doesn't matter.
Associative Properties		
$(a + b) + c = a + (b + c)$	$(2 + 4) + 7 = 2 + (4 + 7)$	When we add three numbers, it doesn't matter which two we add first.
$(ab)c = a(bc)$	$(3 \cdot 7) \cdot 5 = 3 \cdot (7 \cdot 5)$	When we multiply three numbers, it doesn't matter which two we multiply first.
Distributive Property		
$a(b + c) = ab + ac$	$2 \cdot (3 + 5) = 2 \cdot 3 + 2 \cdot 5$	When we multiply a number by a sum of two numbers, we get the same result as we get if we multiply the number by each of the terms and then add the results.
$(b + c)a = ab + ac$	$(3 + 5) \cdot 2 = 2 \cdot 3 + 2 \cdot 5$	

Please read page 3-4 of the textbook.

PROPERTIES OF NEGATIVES

PROPERTIES OF NEGATIVES

Property

Example

1. $(-1)a = -a$

$(-1)5 = -5$

2. $-(-a) = a$

$-(-5) = 5$

3. $(-a)b = a(-b) = -(ab)$

$(-5)7 = 5(-7) = -(5 \cdot 7)$

4. $(-a)(-b) = ab$

$(-4)(-3) = 4 \cdot 3$

5. $-(a + b) = -a - b$

$-(3 + 5) = -3 - 5$

6. $-(a - b) = b - a$

$-(5 - 8) = 8 - 5$

Please read page 4 of the textbook.

PROPERTIES OF FRACTIONS

PROPERTIES OF FRACTIONS

Property

$$1. \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

$$2. \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

$$3. \frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$$

$$4. \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$

$$5. \frac{ac}{bc} = \frac{a}{b}$$

$$6. \text{ If } \frac{a}{b} = \frac{c}{d}, \text{ then } ad = bc$$

Example

$$\frac{2}{3} \cdot \frac{5}{7} = \frac{2 \cdot 5}{3 \cdot 7} = \frac{10}{21}$$

$$\frac{2}{3} \div \frac{5}{7} = \frac{2}{3} \cdot \frac{7}{5} = \frac{14}{15}$$

$$\frac{2}{5} + \frac{7}{5} = \frac{2+7}{5} = \frac{9}{5}$$

$$\frac{2}{5} + \frac{3}{7} = \frac{2 \cdot 7 + 3 \cdot 5}{35} = \frac{29}{35}$$

$$\frac{2 \cdot 5}{3 \cdot 5} = \frac{2}{3}$$

$$\frac{2}{3} = \frac{6}{9}, \text{ so } 2 \cdot 9 = 3 \cdot 6$$

Description

When **multiplying fractions**, multiply numerators and denominators.

When **dividing fractions**, invert the divisor and multiply.

When **adding fractions** with the **same denominator**, add the numerators.

When **adding fractions** with **different denominators**, find a common denominator. Then add the numerators.

Cancel numbers that are **common factors** in numerator and denominator.

Cross-multiply.

Please read page 5 of the textbook.

INTEGER EXPONENTS

DEFINITION:

If a is any real number and n is a **positive** integer, then the **n th power** of a is

$$a^n := \underbrace{a \times a \times \dots \times a}_{n \text{ factors}}$$

The number a is called the **base**, and n is called the **exponent**. Furthermore, if a is a **non-zero** real number, then

$$a^0 := 1 \text{ and } a^{-n} := \frac{1}{a^n}.$$

E.g.:

- $(0.1)^3 = 0.1 \times 0.1 \times 0.1 = 0.001.$
- $(-7)^2 = (-7) \times (-7) = 49.$
- $-7^2 = -(7^2) = -(7 \times 7) = -49.$
- $\left(\frac{\pi}{2}\right)^0 = 1.$
- $(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}.$

ORDER OF OPERATIONS

⚠ Recall the convention for the order of performing operations

PEMDAS (with descending priority):

- **P**arentheses (incl. absolute values, brackets, fraction lines, radicals)
- **E**xponents
- **M**ultiplications and **D**ivisions (the order is from left to right of the expression)
- **A**dditions and **S**ubtractions (the order is from left to right of the expression)

E.g.:

- That's why we perform the **exponential** first in -7^2 , followed by the **multiplication** by -1 .
- $3 - 4 \div 2 = 3 - (4 \div 2) = 3 - 2 = 1$.
- $3 - 2 + 1 = (3 - 2) + 1 = 2$ and $3 \div 3 \times 2 = (3 \div 3) \times 2 = 2$, using the left to right convention.

In fact, we'll always use parentheses as much as we can to avoid ambiguity, and so should you.

LAWS OF EXPONENTS

LAWS OF EXPONENTS

Law	Example	Description
1. $a^m a^n = a^{m+n}$	$3^2 \cdot 3^5 = 3^{2+5} = 3^7$	To multiply two powers of the same number, add the exponents.
2. $\frac{a^m}{a^n} = a^{m-n}$	$\frac{3^5}{3^2} = 3^{5-2} = 3^3$	To divide two powers of the same number, subtract the exponents.
3. $(a^m)^n = a^{mn}$	$(3^2)^5 = 3^{2 \cdot 5} = 3^{10}$	To raise a power to a new power, multiply the exponents.
4. $(ab)^n = a^n b^n$	$(3 \cdot 4)^2 = 3^2 \cdot 4^2$	To raise a product to a power, raise each factor to the power.
5. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	$\left(\frac{3}{4}\right)^2 = \frac{3^2}{4^2}$	To raise a quotient to a power, raise both numerator and denominator to the power.
6. $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$	$\left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2$	To raise a fraction to a negative power, invert the fraction and change the sign of the exponent.
7. $\frac{a^{-n}}{b^{-m}} = \frac{b^m}{a^n}$	$\frac{3^{-2}}{4^{-5}} = \frac{4^5}{3^2}$	To move a number raised to a power from numerator to denominator or from denominator to numerator, change the sign of the exponent.

The principle behind these laws is to count the number of times the base appears as a factor. They are also true for rational exponents, which we'll see next.

ACTIVITY 6

Simplify each expression, and eliminate any negative exponents.

(a) $w^{-2}w^{-4}w^5$

(b) $\frac{x^{16}}{x^{10}}$

(c) $(2a^3a^2)^4$

(d) $(-3z^2)^3(2z^3)$

(e) $(8m^2n^4)\left(\frac{1}{2}n^{-2}\right)$

(f) $(3a^4b^{-2})^3(a^2b^{-1})$

(g) $\left(\frac{a^2}{b}\right)^5\left(\frac{a^3b^2}{c^3}\right)^3$

(h) $\frac{(u^{-1}v^2)^2}{(u^3v^{-2})^3}$

(i) $\left(\frac{s^2t^{-4}}{5s^{-1}t}\right)^{-2}$

ACTIVITY 6: SOLUTION

$$(a) w^{-2}w^{-4}w^5 = w^{-2-4+5} = w^{-1} = \frac{1}{w}.$$

$$(b) \frac{x^{16}}{x^{10}} = x^{16-10} = x^6.$$

$$(c) (2a^3a^2)^4 = (2a^{3+2})^4 = (2a^5)^4 = 2^4a^{5 \times 4} = 16a^{20}.$$

$$(d) (-3z^2)^3 (2z^3) = (-3)^3 z^{2 \times 3} \times 2z^3 = -27 \times 2 \times z^{6+3} = -54z^9.$$

$$(e) (8m^2n^4) \left(\frac{1}{2}n^{-2}\right) = \frac{8}{2}m^2n^{4-2} = 4m^2n^2.$$

$$(f) (3a^4b^{-2})^3 (a^2b^{-1}) = 3^3a^{4 \times 3}b^{(-2) \times 3} \times a^2b^{-1} = 27a^{12+2}b^{-6-1} = 27\frac{a^{14}}{b^7}.$$

$$(g) \left(\frac{a^2}{b}\right)^5 \left(\frac{a^3b^2}{c^3}\right)^3 = \frac{a^{10}}{b^5} \times \frac{a^{3 \times 3}b^{2 \times 3}}{c^{3 \times 3}} = \frac{a^{10+9}b^6}{b^5c^9} = \frac{a^{19}b}{c^9}.$$

$$(h) \frac{(u^{-1}v^2)^2}{(u^3v^{-2})^3} = \frac{u^{-2}v^4}{u^9v^{-6}} = \frac{v^4v^6}{u^9u^2} = \frac{v^{10}}{u^{11}}.$$

$$(i) \left(\frac{s^2t^{-4}}{5s^{-1}t}\right)^{-2} = \frac{s^{-4}t^8}{5^{-2}s^2t^{-2}} = \frac{5^2t^8t^2}{s^4s^2} = \frac{25t^{10}}{s^6}.$$

RADICALS

DEFINITION:

If n is any positive integer, then the **principal n th root** of a , denoted as $a^{\frac{1}{n}}$ or $\sqrt[n]{a}$ (called a **radical**), is defined as

$$a^{\frac{1}{n}} = \sqrt[n]{a} := b, \text{ where } b^n = a.$$

If n is even, we must have both $a, b \geq 0$. (The word principal corresponds to this choice among all possible roots.)

Note that this definition agrees with the convention that $\sqrt{x}$ is the **positive** square root of x .

E.g.:

- $\sqrt[3]{8} = 2$, because $2^3 = 8$.
- $\sqrt[3]{-8} = -2$, because $(-2)^3 = -8$.
- $\sqrt[4]{81} = 3$, because $3 \geq 0$ (as 4 is even) and $3^4 = 81$.
- $\sqrt[4]{-81}$ is undefined in $\mathbb{R}$, we need complex numbers to handle this, which is beyond our syllabus.

PROPERTIES OF n TH ROOTS

PROPERTIES OF n th ROOTS

Property

Example

$$1. \sqrt[n]{ab} = \sqrt[n]{a}\sqrt[n]{b}$$

$$\sqrt[3]{-8 \cdot 27} = \sqrt[3]{-8}\sqrt[3]{27} = (-2)(3) = -6$$

$$2. \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\sqrt[4]{\frac{16}{81}} = \frac{\sqrt[4]{16}}{\sqrt[4]{81}} = \frac{2}{3}$$

$$3. \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

$$\sqrt{\sqrt[3]{729}} = \sqrt[6]{729} = 3$$

$$4. \sqrt[n]{a^n} = a \quad \text{if } n \text{ is odd}$$

$$\sqrt[3]{(-5)^3} = -5, \quad \sqrt[5]{2^5} = 2$$

$$5. \sqrt[n]{a^n} = |a| \quad \text{if } n \text{ is even}$$

$$\sqrt[4]{(-3)^4} = |-3| = 3$$

⚠ $\sqrt{x^2} \neq x$, but instead $\sqrt{x^2} = |x|$.

RATIONAL EXPONENTS

DEFINITION:

For any rational exponent $\frac{m}{n}$ in lowest terms, where m and n are integers and $n \neq 0$, we define

$$a^{\frac{m}{n}} := (\sqrt[n]{a})^m \text{ or equivalently } a^{\frac{m}{n}} := \sqrt[n]{a^m}.$$

If n is even, then we require $a > 0$.

E.g.:

- $(-8)^{\frac{2}{3}} = (\sqrt[3]{-8})^2 = (-2)^2 = 4.$
- $125^{-\frac{4}{3}} = \frac{1}{125^{\frac{4}{3}}} = \frac{1}{(\sqrt[3]{125})^4} = \frac{1}{5^4} = \frac{1}{625}.$

The laws of exponents on page 62 still hold for rational exponents. For instance, the reason we denote $a^{\frac{1}{n}} = \sqrt[n]{a}$ is because

$$a = (\sqrt[n]{a})^n = \left(a^{\frac{1}{n}}\right)^n = a^{\frac{n}{n}} = a^1 = a.$$

ACTIVITY 7

Simplify each expression. The letters denote positive real numbers.

(a) $\sqrt[6]{64a^6b^7}$

(b) $\sqrt[3]{a^2b}\sqrt[3]{64a^4b}$

(c) $\sqrt[3]{\sqrt{64x^6}}$

(d) $\sqrt{9a} + \sqrt{16a}$

(e) $\sqrt{32} + \sqrt{18}$

(f) $\sqrt{81x^2 + 81}$

(g) $32^{\frac{2}{5}}$

(h) $-8^{\frac{1}{3}}$

(i) $(-8)^{\frac{1}{3}}$

(j) $x^{\frac{3}{4}}x^{\frac{5}{4}}$

(k) $\frac{w^{\frac{4}{3}}w^{\frac{2}{3}}}{w^{\frac{1}{3}}}$

(l) $\left(\frac{x^{\frac{3}{2}}}{y^{-\frac{1}{2}}}\right)^4 \left(\frac{x^{-2}}{y^3}\right)$

(m) $\left(\frac{4y^3z^{\frac{2}{3}}}{x^{\frac{1}{2}}}\right)^2 \left(\frac{x^{-3}y^6}{8z^4}\right)^{\frac{1}{3}}$

(n) $\sqrt[6]{y^5}\sqrt[3]{y^2}$

(o) $(5\sqrt[3]{x})(2\sqrt[4]{x})$

(p) $\sqrt[3]{y\sqrt{y}}$

(q) $\sqrt{\frac{16u^3v}{uv^5}}$

ACTIVITY 7: SOLUTION

$$(a) \sqrt[6]{64a^6b^7} = (2^6a^6b^7)^{\frac{1}{6}} = 2^{\frac{6}{6}}a^{\frac{6}{6}}b^{\frac{7}{6}} = 2ab^{\frac{7}{6}}.$$

$$(b) \sqrt[3]{a^2b}\sqrt[3]{64a^4b} = \sqrt[3]{(a^2b)(64a^4b)} = \sqrt[3]{(2^6a^6b^2)} = (2^6a^6b^2)^{\frac{1}{3}} = 2^{\frac{6}{3}}a^{\frac{6}{3}}b^{\frac{2}{3}} = 4a^2b^{\frac{2}{3}}.$$

$$(c) \sqrt[3]{\sqrt{64x^6}} = \left((2^6x^6)^{\frac{1}{2}}\right)^{\frac{1}{3}} = (2^6x^6)^{\frac{1}{2} \times \frac{1}{3}} = (2^6x^6)^{\frac{1}{6}} = 2x.$$

$$(d) \sqrt{9a} + \sqrt{16a} = \sqrt{9}\sqrt{a} + \sqrt{16}\sqrt{a} = 3\sqrt{a} + 4\sqrt{a} = 7\sqrt{a}.$$

$$(e) \sqrt{32} + \sqrt{18} = \sqrt{16 \times 2} + \sqrt{9 \times 2} = \sqrt{16}\sqrt{2} + \sqrt{9}\sqrt{2} = 4\sqrt{2} + 3\sqrt{2}.$$

$$(f) \sqrt{81x^2 + 81} = \sqrt{81(x^2 + 1)} = \sqrt{81}\sqrt{x^2 + 1} = 9\sqrt{x^2 + 1}.$$

$$(g) 32^{\frac{2}{5}} = (2^5)^{\frac{2}{5}} = 2^{\frac{5 \times 2}{5}} = 2^2 = 4.$$

$$(h) -8^{\frac{1}{3}} = -(8^{\frac{1}{3}}) = -(2) = -2.$$

$$(i) (-8)^{\frac{1}{3}} = ((-2)^3)^{\frac{1}{3}} = -2.$$

$$(j) x^{\frac{3}{4}}x^{\frac{5}{4}} = x^{\frac{3}{4} + \frac{5}{4}} = x^{\frac{8}{4}} = x^2.$$

ACTIVITY 7: SOLUTION

$$(k) \frac{w^{\frac{4}{3}} w^{\frac{2}{3}}}{w^{\frac{1}{3}}} = w^{\frac{4}{3} + \frac{2}{3} - \frac{1}{3}} = w^{\frac{5}{3}}.$$

$$(l) \left(\frac{x^{\frac{3}{2}}}{y^{-\frac{1}{2}}} \right)^4 \left(\frac{x^{-2}}{y^3} \right) = \frac{x^{\frac{12}{2}}}{y^{-\frac{4}{2}}} \frac{1}{x^2 y^3} = \frac{x^{6-2}}{y^{3-2}} = \frac{x^4}{y}.$$

$$(m) \left(\frac{4y^3 z^{\frac{2}{3}}}{x^{\frac{1}{2}}} \right)^2 \left(\frac{x^{-3} y^6}{8z^4} \right)^{\frac{1}{3}} = \frac{4^2 y^6 z^{\frac{4}{3}}}{x} \frac{x^{-1} y^2}{8^{\frac{1}{3}} z^{\frac{4}{3}}} = \frac{16 y^6 z^{\frac{4}{3}}}{x} \frac{y^2}{2 x z^{\frac{4}{3}}} = \frac{8 y^8}{x^2}.$$

$$(n) \sqrt[6]{y^5} \sqrt[3]{y^2} = y^{\frac{5}{6}} y^{\frac{2}{3}} = y^{\frac{5}{6} + \frac{2}{3}} = y^{\frac{9}{6}} = y^{\frac{3}{2}}.$$

$$(o) (5\sqrt[3]{x})(2\sqrt[4]{x}) = 10x^{\frac{1}{3}} x^{\frac{1}{4}} = 10x^{\frac{1}{3} + \frac{1}{4}} = 10x^{\frac{7}{12}}.$$

$$(p) \sqrt[3]{y\sqrt{y}} = \left(yy^{\frac{1}{2}} \right)^{\frac{1}{3}} = \left(y^{\frac{3}{2}} \right)^{\frac{1}{3}} = y^{\frac{3}{2} \times \frac{1}{3}} = y^{\frac{1}{2}}.$$

$$(q) \sqrt{\frac{16u^3v}{uv^5}} = \left(\frac{16u^{3-1}}{v^{5-1}} \right)^{\frac{1}{2}} = \frac{16^{\frac{1}{2}} u^{\frac{2}{2}}}{v^{\frac{4}{2}}} = \frac{4u}{v^2}.$$